

that are potentially associated with the outcome evolution of $Y(0)$ during post-treatment periods. We explicitly rule out incorporating post-treatment covariates as they can potentially be affected by the treatment; see, e.g., [Wooldridge \(2005b\)](#), for a related discussion under the unconfoundedness setup.

3 Summarizing Group-Time Average Treatment Effects

The previous section shows that we can identify the $ATT(g, t)$'s by restricting treatment anticipation behavior and imposing a conditional parallel trends assumption. In many applications, the $ATT(g, t)$'s can be the ultimate causal parameters of interest. They can be used to highlight treatment effect heterogeneity across different groups g , at different points in time t , and across different lengths of treatment exposure, $e = t - g$. In other situations, however, researchers may want to combine these different $ATT(g, t)$'s to form more aggregated causal parameters. For instance, if the number of groups and time periods is relatively large, it may be challenging to interpret many group-time average treatment effects.

In this section, we consider different aggregation schemes for the $ATT(g, t)$'s that allow researchers to form a variety of summary measures of the causal effects of a given policy. Our aggregation schemes are of the form

$$\theta = \sum_{g \in \mathcal{G}} \sum_{t=2}^{\tau} w(g, t) \cdot ATT(g, t), \quad (3.1)$$

where $w(g, t)$ are carefully-chosen (known or estimable) weighting functions specified by the researcher such that θ can be used to address a well-posed empirical/policy question. Different choices of $w(g, t)$ allows researchers to highlight different types of treatment effect heterogeneity. We pay particular attention to aggregations that result in a single overall treatment effect summary parameter as well as to aggregations related to understanding dynamic effects as is commonly done in event-study analysis. Of course, many other aggregated parameters of the type (3.1) can be easily constructed following our framework. We illustrate this point by also summarizing heterogeneity with respect to group or by calendar time.

Before proceeding with the discussion on how to construct these different aggregated parameters, it is worth revisiting the two most popular treatment effect summary measures used by practitioners in DiD setups. These are based on the “static” and “dynamic” two-way fixed effects (TWFE) linear regression specifications

$$Y_{i,t} = \alpha_t + \alpha_g + \beta D_{i,t} + \epsilon_{i,t}, \quad (3.2)$$

$$Y_{i,t} = \alpha_t + \alpha_g + \sum_{e=-K}^{-2} \delta_e^{anticip} \cdot D_{i,t}^e + \sum_{e=0}^L \beta_e \cdot D_{i,t}^e + v_{i,t}, \quad (3.3)$$

respectively, where α_t is a time fixed effect, α_g is a group fixed effect, $\epsilon_{i,t}$ and $v_{i,t}$ are error terms, $D_{i,t}^e = 1\{t - G_i = e\}$ is an indicator for unit i being e periods away from initial treatment at time t , and K and L are positive constants. The parameter of interest in the static TWFE specification is β , which, in applications, is typically interpreted as an overall effect of participating in the treatment across groups and time periods. In the dynamic TWFE specification, practitioners usually focus on the β_e , $e \geq 0$, and these parameters are typically interpreted as measuring the effect of participating in the

treatment at different lengths of exposure to the treatment.

Despite the popularity of these specifications, recent research has shown that one must be very careful in attaching a causal interpretation to these aggregated parameters. For instance, [Borusyak and Jaravel \(2017\)](#), [Goodman-Bacon \(2019\)](#), [de Chaisemartin and D’Haultfœuille \(2020\)](#), and [Athey and Imbens \(2018\)](#) have shown that, in general, β recovers a weighted average of some underlying treatment effect parameters but some of the weights on these parameters can be negative. This can potentially lead to particularly problematic cases such as the effect of the treatment being positive for all units, but the TWFE estimation resulting in estimates of β that are negative. Even in cases where the weights are not negative, the weights on underlying treatment effect parameters are entirely driven by the TWFE estimation strategy and are sensitive to the size of each group, the timing of treatment, and the total number of time periods (see Theorem 1 in [Goodman-Bacon \(2019\)](#)). The results in this section can be used in exactly the same setup to identify a single interpretable average treatment effect parameter and, thus, provide a way to circumvent the issues with the more common approach.

As discussed by [Goodman-Bacon \(2019\)](#), the “negative weight problem” associated with β arises when treatment effects evolve over time. Thus, one may wonder if such problems would still be present when considering more general, dynamic specifications such as (3.3). [Sun and Abraham \(2020\)](#) shows that this is still the case as the β'_e s associated with (3.3) do not recover easy-to-interpret causal parameters and still generally suffer from the same sorts of “negative weighting problems.” In contrast to this, we provide a simple way to directly aggregate our group-time average treatment effects into average treatment effects across different lengths of exposure to the treatment.

3.1 Aggregations to Highlight Treatment Effect Heterogeneity

Next, we discuss several partial aggregations of the group-time average treatment effects in order to summarize different dimensions of treatment effect heterogeneity. Although there are additional possibilities, we focus our discussion below on how to answer three particular questions: (a) How does the effect of participating in the treatment vary with length of exposure to the treatment? (b) Do groups that are treated earlier have, on average, higher/lower average treatment effects than groups that are treated later? (c) What is the cumulative average treatment effect of the policy across all groups until some particular point in time? Throughout this section, to avoid notation clutter, we assume that units do not anticipate treatment, i.e., we consider the case where Assumption 3 holds with $\delta = 0$. We also assume that a “never treated” group is available.

3.1.1 How do average treatment effects vary with length of exposure to the treatment?

One of the most popular questions that arises in DiD setups with multiple time periods concerns treatment effect dynamics: How does the effect of participating in the treatment vary with length of exposure to the treatment? For instance, do average treatment effects increase/decrease with elapsed treatment time? Indeed, answering this type of question is often the main motivation for using the event study regression in (3.3), though, as we mentioned above, that sort of regression may not be suitable for such a task. In this section, we propose an aggregation scheme that is suitable to highlight treatment effect heterogeneity with respect to length of exposure to the treatment that does not suffer from the drawbacks associated

Table 1: Weights on $ATT(g, t)$ for Aggregated Parameters

Parameter	$w(g, t)$
$\theta_{es}(e)$	$w_e^{es}(g, t) = \mathbf{1}\{g + e \leq \mathcal{T}\} \mathbf{1}\{t - g = e\} P(G = g G + e \leq \mathcal{T})$
$\theta_{es}^{bal}(e, e')$	$w_e^{es, bal}(g, t) = \mathbf{1}\{g + e' \leq \mathcal{T}\} \mathbf{1}\{t - g = e\} P(G = g G + e' \leq \mathcal{T})$
$\theta_{sel}(\tilde{g})$	$w_{\tilde{g}}^s(g, t) = \mathbf{1}\{t \geq g\} \mathbf{1}\{g = \tilde{g}\} / (\mathcal{T} - g + 1)$
$\theta_c(\tilde{t})$	$w_{\tilde{t}}^c(g, t) = \mathbf{1}\{t \geq g\} \mathbf{1}\{t = \tilde{t}\} P(G = g G \leq t)$
$\theta_c^{cumu}(\tilde{t})$	$w_{\tilde{t}}^{c, cumu}(g, t) = \mathbf{1}\{t \geq g\} \mathbf{1}\{t \leq \tilde{t}\} P(G = g G \leq t)$
θ_W^O	$w_W^O(g, t) = \mathbf{1}\{t \geq g\} P(G = g G \leq \mathcal{T}) / \sum_{g \in \mathcal{G}} \sum_{t=2}^{\mathcal{T}} \mathbf{1}\{t \geq g\} P(G = g G \leq \mathcal{T})$
θ_{sel}^O	$w_{sel}^O(g, t) = \mathbf{1}\{t \geq g\} P(G = g G \leq \mathcal{T}) / (\mathcal{T} - g + 1)$

Notes: This table provides expressions for the weights on each $ATT(g, t)$ (as in Equation 3.1) for each parameter discussed in this section. In all cases except for $\theta_c^{cumu}(\tilde{t})$, the weights are all non-negative and sum to one. For $\theta_c^{cumu}(\tilde{t})$, the weights are all non-negative but sum up to $\tilde{t} - 1$ (rather than one), but this is just a reflection of $\theta_c^{cumu}(\tilde{t})$ being a cumulative treatment effect measure.

with the event study regression in (3.3).

Let e denote event-time, i.e., $e = t - g$ denotes the time elapsed since treatment was adopted. Recall that G denotes the time period that a unit is first treated. Thus, a way to aggregate the $ATT(g, t)$'s to highlight treatment effect heterogeneity with respect to e is

$$\theta_{es}(e) = \sum_{g \in \mathcal{G}} \mathbf{1}\{g + e \leq \mathcal{T}\} P(G = g | G + e \leq \mathcal{T}) ATT(g, g + e). \quad (3.4)$$

This is the average effect of participating in the treatment e time periods after the treatment was adopted across all groups that are ever observed to have participated in the treatment for exactly e time periods. Here, the “on impact” average effect of participating in the treatment occurs for $e = 0$. $\theta_{es}(e)$ is the natural target for event study regressions that are common in applied work, though it completely avoids the pitfalls associated with the dynamic TWFE specification in (3.3).¹⁰

In event study regressions, it is common to plot β_e across different values of e and to interpret differences as being due to treatment effect dynamics. Similarly, one can plot $\theta_{es}(e)$ across different e 's to better understand treatment effect dynamics. When doing so, it is important to be aware that these comparisons may include compositional changes that can complicate the interpretation of these parameters (note that the same complications arise for event study regressions as well). To see this, for

¹⁰Many of the parameters in this section involve expressions that have similar components as the one for $\theta_{es}(e)$ in (3.4), and it is worth mentioning a few extra details for this case that are common to the other expressions below. The term involving the indicator function, $\mathbf{1}\{g + e \leq \mathcal{T}\}$, limits consideration to identified group-time average treatment effects. The summation over groups with group specific weights, in this case given by $P(G = g | G + e \leq \mathcal{T})$, calculates an average, weighted by group size, of $ATT(g, t)$'s that are involved in a particular aggregation. In addition, it is straightforward to show that $\theta_{es}(e)$ can be written in the form of θ in Equation (3.1). Throughout this section, we have written each parameter of interest in its most intuitive form. Weights for each parameter in this section corresponding to the form of the weights in Equation (3.1) are provided in Table 1.

$0 \leq e_1 < e_2 \leq \mathcal{T} - 2$, consider the difference between $\theta_{es}(e_2)$ and $\theta_{es}(e_1)$ which is given by

$$\begin{aligned} \theta_{es}(e_2) - \theta_{es}(e_1) = & \sum_{g \in \mathcal{G}} \mathbf{1}\{g + e_1 \leq \mathcal{T}\} P(G = g | G + e_1 \leq \mathcal{T}) \underbrace{(ATT(g, g + e_2) - ATT(g, g + e_1))}_{\text{dynamic effect for group } g} \quad (3.5) \\ & + \sum_{g \in \mathcal{G}} \mathbf{1}\{g + e_2 \leq \mathcal{T}\} \underbrace{(P(G = g | G + e_2 \leq \mathcal{T}) - P(G = g | G + e_1 \leq \mathcal{T}))}_{\text{differences in weights}} ATT(g, g + e_2) \\ & - \sum_{g \in \mathcal{G}} \underbrace{\mathbf{1}\{\mathcal{T} - e_2 < g \leq \mathcal{T} - e_1\}}_{\text{different composition of groups}} P(G = g | G + e_1 \leq \mathcal{T}) ATT(g, g + e_2). \end{aligned}$$

From the above decomposition it becomes clear that comparing $\theta_{es}(e)$ at two different values of e provides a weighted average of the dynamic effect of participating in the treatment – the first component on the right-hand side of (3.5) – plus two extra undesirable terms. Both of these undesirable terms are due to different compositions of groups at different event times.¹¹ The first term arises because the weights at each length of exposure differ due to the changing composition of groups at each event time. The second term comes directly from different compositions of groups at each length of exposure. These two additional terms may prevent one from interpreting the differences in $\theta_{es}(e)$ across different values of e as being actual dynamic effects of participating in the treatment unless one is willing to impose that $ATT(g, g + e)$ does not vary with g for any $e \geq 0$; i.e., that dynamic effects are common across groups.¹² However, this sort of homogeneity condition may be deemed too strong in many applications.

A simple alternative causal parameter that can be used to highlight treatment effect dynamics with respect to length of exposure to the treatment and does not suffer from the issue of compositional changes highlighted in (3.5) arises from “balancing” the groups with respect to event time, i.e., to only aggregate the $ATT(g, t)$ ’s for a fixed set of groups that are exposed to the treatment for at least some particular number of time periods and thereby circumvent the issue of compositional changes across different values of e . In particular, for some event time e' with $0 \leq e \leq e' \leq \mathcal{T} - 2$, let

$$\theta_{es}^{bal}(e; e') = \sum_{g \in \mathcal{G}} \mathbf{1}\{g + e' \leq \mathcal{T}\} ATT(g, g + e) P(G = g | G + e' \leq \mathcal{T}). \quad (3.6)$$

Notice that the definition of $\theta_{es}^{bal}(e; e')$ is very similar to $\theta_{es}(e)$ except that it calculates the average group-time average treatment effect for units whose event time is equal to e *and who are observed to participate in the treatment for at least e' periods*. In this case, since the composition of groups is the same across all values of e , the additional terms in (3.5) do not show up at all and differences in $\theta_{es}^{bal}(e; e')$ across different values of e cannot be due to differences in the composition of groups at different values of e . As an example, when one is interested in analyzing the evolution of treatment effects up to 5 periods after treatment was implemented, one can set $e' = 5$ and, this way, the same groups of units will be used when computing $\theta_{es}^{bal}(0; 5), \theta_{es}^{bal}(1; 5), \dots, \theta_{es}^{bal}(5; 5)$.

The price one pays for “balancing” the groups with respect to event time is that fewer groups are used to compute these event-study-type estimands, which can lead to less informative inference. Thus, in

¹¹The composition changes mentioned here arise due to the staggered adoption of the treatment. For example, when $\mathcal{T} = 3$, groups 2 and 3 both show up in the expression for $\theta_{es}(0)$, but only group 2 shows up in the expression for $\theta_{es}(1)$.

¹²If $ATT(g, g + e)$ does not vary with g for any $e \geq 0$, it is straightforward to show that the last two terms of (3.5) sum up to 0.